

An increase of *NPY1* expression leads to inhibitory phosphorylation of PIN-FORMED (PIN) proteins and suppression of *pinoid* (*pid*) null mutants

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eLife Assessment

This **valuable** study concerns a highly interesting and biologically relevant topic, the regulation of the PIN auxin transporter, which is of broad interest to the plant biology community. The authors propose NPY1 to act downstream of PID in auxin-mediated development by modulating PIN phosphorylation, which, if experimentally solidified, would expand our understanding of PIN regulation. While the genetic evidence is **solid**, the mechanistic role of NPY1 and the functional relevance of phosphorylated PIN residues are still uncertain. There are also concerns regarding experimental rigor and methodological transparency.

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Abstract

The PINOID (PID) protein kinase is required for flower initiation in Arabidopsis. The *pid* mutants fail to initiate flowers on inflorescences, a phenotype that is mimicked by disrupting either the *NAKED PINS IN YUC MUTANTS* (*NPY*) gene family or *PIN FORMED 1* (*PIN1*). Both PID and NPY1 have been reported to positively modulate PIN-mediated polar auxin transport. Here, we show that overexpression of *NPY1* (*NPY1 OE*) completely suppressed *pid* null mutants, demonstrating that *NPY1* functions downstream of *PID*. *NPY1 OE* triggered phosphorylation of PIN proteins at multiple sites that are mostly different from the previously characterized phosphorylation sites regardless of the presence of *PID*. Phosphorylation of the newly identified PIN sites in *NPY1 OE* plants likely leads to the inhibition of PIN functions, as we previously showed that *pid* is suppressed by decreasing *PIN1* gene dosage or decreasing PIN1 activity. Furthermore, we show that the Ser/Thr rich C-terminal motif in NPY1 is phosphorylated and is required for *pid* suppression by *NPY1 OE*. Overexpression of *NPY1* that lacked the C-terminal motif (*NPY1ΔC*) failed to rescue *pid*, but overexpression of *NPY1ΔC* was still able to trigger phosphorylation of PIN proteins including PIN2, which is known to cause agravitropic roots when mutated. *NPY1ΔC* overexpression plants displayed a complete loss of root gravitropic response, likely caused by PIN2

phosphorylation. Our results suggest a pathway for auxin mediated-flower initiation, in which PID regulates NPY1 accumulation and/or activity, and subsequently, NPY1 triggers phosphorylation of PIN proteins and inhibition of PIN functions.

Introduction

Auxin is required for flower initiation in *Arabidopsis* and other plants (Cheng and Zhao, 2007; Yamaguchi et al., 2013). Several *Arabidopsis* mutants and mutant combinations fail to initiate flowers on inflorescences, despite the fact that they can undergo the normal transition from vegetative growth to reproductive growth, resulting in the formation of pin-like inflorescences (Bennett et al., 1995; Przemeck et al., 1996; Galweiler et al., 1998; Cheng et al., 2007). All of the known pin-like *Arabidopsis* mutants are caused by defects in some aspects of auxin biology. *PIN-FORMED 1* (*PIN1*) was the first characterized gene that causes the formation of pin-like inflorescences when compromised (Galweiler et al., 1998). *PIN1* is an auxin efflux carrier that transports auxin from the cytosol into the extracellular space (Yang et al., 2022). Plants grown on media containing the polar auxin transport inhibitor N-1-naphthylphthalamic acid (NPA) phenocopy *pin1* mutants (Okada et al., 1991). It has been shown that NPA can directly bind to *PIN1* to inhibit auxin transport (Yang et al., 2022). Disruption of *PINOID* (*PID*), which encodes a Ser/Thr protein kinase, also leads to the formation of pin-like inflorescences (Smyth and Alvarez, 1995; Christensen et al., 2000). Overexpression of *PID* leads to phenotypes similar to the well-characterized auxin resistant mutants, linking *PID* functions to auxin (Christensen et al., 2000). *NAKED PINS IN YUC MUTANTS 1* (*NPY1*) (Cheng et al., 2007), which is also called *ENHANCER OF PINOID* (*ENP*) (Trembl et al., 2005) and *MACCHI-BOU 4* (*MAB4*) (Furutani et al., 2007), was isolated from genetic screens for *pid* enhancers and for enhancers of *yuc1 yuc4* double mutants (Cheng et al., 2006), which are defective in auxin biosynthesis (Cheng et al., 2007). The *yuc1 yuc4 npy1* triple mutants produce pin-like inflorescences similar to those in *pin1* and *pid* (Cheng et al., 2007). Inactivation of *NPY1* and its close homologs *NPY3* and *NPY5* also causes the failure of floral initiation from inflorescences (Cheng et al., 2008). The *yuc1 yuc2 yuc4 yuc6* quadruple mutants also develop small pin-like structures (Cheng et al., 2006).

Although the aforementioned genes are known to participate in auxin-mediated flower development, the mechanisms by which the genes regulate floral development and the relationships between them are not fully understood. The predominant model regarding the relationship among *PIN1*, *PID*, and *NPY* genes is centered on regulating PIN polarity, localization, and activity by *PID* and *NPY* proteins. It was reported that *PID* directly phosphorylates the hydrophilic loop of PIN proteins, resulting in changes in PIN polarity and activation of PIN-mediated auxin export (Friml et al., 2004; Michniewicz et al., 2007). *NPYs* were suggested to regulate PIN internalization through the endocytosis pathway (Furutani et al., 2011). Recently, it was reported that *NPY1*, *PID*, and *PIN1* form a protein complex at the plasma membrane and that the recruitment of *NPYs* to the plasma-membrane by PIN limits the lateral diffusion of PINs, thus maintaining PIN polarity (Glanc et al., 2021). Phosphorylation of PIN by *PID* enhances the recruitment of *NPY* to plasma membrane, which then promotes PIN phosphorylation by recruiting or interacting with AGC kinases, thus forming a self-reinforcing loop to maintain PIN polarity (Glanc et al., 2021). Other protein kinases including D6 PROTEIN KINASE are also involved in phosphorylating PIN proteins and activating PIN-mediated auxin transport (Willige et al., 2013; Barbosa et al., 2014).

Recently, we discovered that *pid* null mutant phenotypes were suppressed when one copy of the *PIN1* gene is inactivated (heterozygous *pin1* mutants) (Mudgett et al., 2023). The *pid* null mutants were also suppressed by the *PIN1*-GFPHDR fusion in which *GFP* was precisely inserted into the *PIN1* gene via CRISPR/Cas9-mediated homologous recombination (Mudgett et al., 2023). *PIN1*-GFPHDR is less active than wild type (WT) *PIN1*, indicating that *pid* is suppressed by lowering either *PIN1* activity or *PIN1* gene dosage. This new type of genetic interaction between *pid* and

pin1, called haplo-complementation, renders *PID* unnecessary for flower development when expressing only 50% of *PIN1*, whereas the presence of 0% or 100% *PIN1* makes *PID* essential. The observed genetic interaction between *pid* and *pin1* suggests a much more complex relationship between the two genes than the current model of *PIN1* phosphorylation by *PID* directly (Mudgett et al., 2023 [DOI](#)). Moreover, it was known that *pid* mutants produce more cotyledons and true leaves than wild type plants whereas *pin1* mutants make fewer cotyledons and fewer true leaves compared to wild type plants (Bennett et al., 1995 [DOI](#)). The opposite cotyledon/true leaf phenotypes of *pid* and *pin1* cannot be accounted for by *PID*-mediated direct phosphorylation/activation of *PIN1*.

In this study, we analyzed the dosage effects of *NPY1* on *pid* phenotypes. We show that overexpression of *NPY1* (*NPY1 OE*) completely suppressed *pid* null mutants, demonstrating that *NPY1* and *PIN1* have opposite dosage effects on *pid* mutants. Furthermore, we find that *NPY1 OE* led to an increase of *PIN* phosphorylation, suggesting that *NPY1 OE*-triggered *PIN* phosphorylation actually inhibits *PIN* functions. Moreover, we discovered that the C-terminal motif of *NPY1*, which contains 30 amino acid residues with 50% Ser/Thr, is required for suppression of *pid* by *NPY1 OE*. Overexpression of *NPY1* that lacked the C-terminal motif (*NPY1ΔC*) did not suppress *pid*, but caused auxin resistance and a complete loss of gravitropism. Overexpression of *NPY1ΔC* also increased phosphorylation of *PIN* proteins including *PIN2*, which causes agravitropic roots when mutated, further suggesting that phosphorylation of certain residues/regions of *PIN* proteins inhibits *PIN* functions. Our results demonstrated that *NPY1* functions downstream of *PID* during flower development. In addition, our results indicate that *NPY1* affects *PIN* phosphorylation, which inhibits *PIN* function and which may account for the suppression of *pid* by *NPY1 OE*.

Results

Protein null *pin1* mutants further confirm haplo-complementation of *pid* mutants by *pin1*

We previously showed that *pid* null mutants were rescued by either *PIN1*-GFPHDR fusion or the presence of only one copy of functional *PIN1* gene, suggesting that a reduction of *PIN1* activity or *PIN1* gene dosage is sufficient to restore the fertility of *pid* (Mudgett et al., 2023 [DOI](#)). Because the *pin1* mutants in our previous study had the potential to produce truncated *PIN1* proteins, we could not completely rule out the possibility that the predicted truncated *PIN1* proteins might have played a role in rescuing *pid* (Mudgett et al., 2023 [DOI](#)). To further clarify the mechanisms of rescuing *pid* by heterozygous *pin1* mutants, we used CRISPR/Cas9 to delete the coding region of *PIN1* so that no truncated *PIN1* proteins would be produced (Figure 1A [DOI](#)). We obtained two *pin1* mutants, which lacked almost the entire coding region of *PIN1* and which were unlikely to produce any truncated *PIN1* proteins (Figure 1A [DOI](#)). The two *pin1* mutants displayed strong *pin*-like phenotypes (Figure 1B [DOI](#) & 1C). In the heterozygous *pin1* mutant backgrounds, the *pid* null mutants no longer produced *pin*-like inflorescences and were able to set seeds (Figure 1B [DOI](#) & 1C). Rescuing *pid* by heterozygous *pin1* mutants was not specific to a particular *pid* allele. Both T-DNA insertion (Figure 1B [DOI](#)) and CRISPR deletion *pid* mutants (Figure 1C [DOI](#)), which were previously described as *pid* null mutants (Mudgett et al., 2023 [DOI](#)), were rescued by the heterozygous *pin1* full deletion mutants (Figure 1 [DOI](#)), demonstrating that *pid* mutants were rescued by a reduction of *PIN1* gene dosage and that suppression of *pid* mutant phenotypes by heterozygous *pin1* mutants in our previous study was not caused by the predicted truncated *PIN1* proteins.

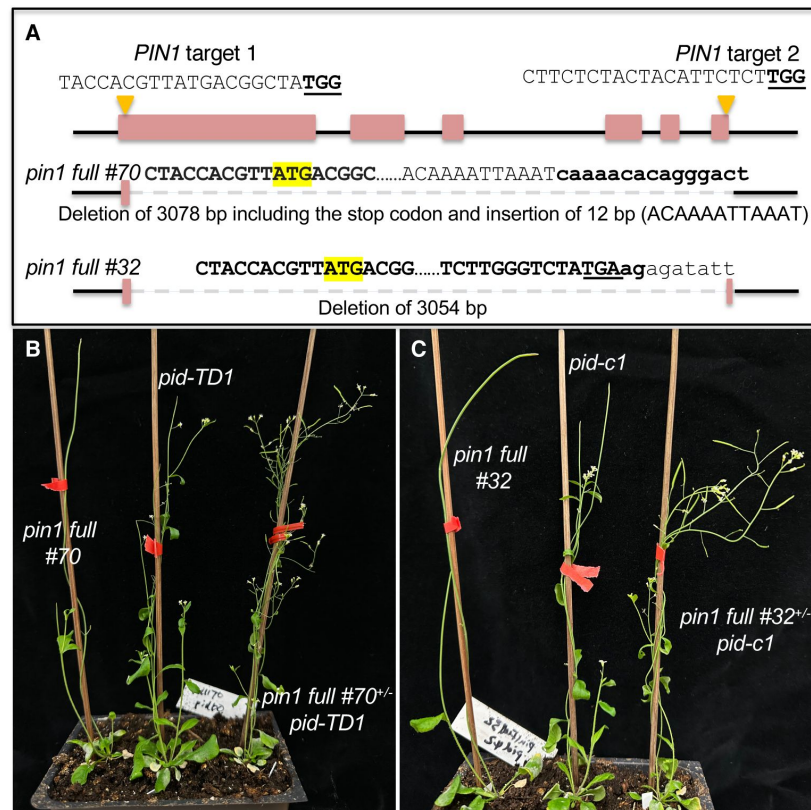


Figure 1.

A complete deletion of one copy of the *PIN1* gene rescues *pid* null mutants.

A) *PIN1* gene is deleted using CRISPR/Cas9 gene editing technology. The two guide RNA target sites are shown with PAM underlined and in bold. The *PIN1* start codon ATG is highlighted in yellow. The sequences flanking the deletions are shown. The two mutants have less than 18 bp of *PIN1* coding sequence and are unlikely to produce any *PIN1* proteins. B) Heterozygous *pin1* full #70, which had a deletion of 3078 bp of the *PIN1* gene, restored the fertility of *pid-TD1*, a T-DNA insertion mutant that is completely sterile in WT *PIN1* background. C) *pin1* full #32, when heterozygous, rescued *pid-c1*, a null *pid* allele.

NPY1 and PIN1 have different effects on *pid* mutants

- a reduction of NPY1 gene dosage cannot rescue *pid*

Inactivation of *NPY1* and its close homologs *NPY3* and *NPY5* leads to the development of pin-like inflorescences (Cheng et al., 2008 [DOI](#)). Moreover, *np1 pid* double mutants fail to develop any cotyledons, a phenotype that was also observed in *pin1 pid* double mutants (Furutani et al., 2004 [DOI](#); Trembl et al., 2005 [DOI](#); Furutani et al., 2007 [DOI](#); Cheng et al., 2008 [DOI](#)). Therefore, we hypothesized that a reduction of *NPY1* gene dosage might also be able to rescue *pid* mutants. We genotyped F2 plants from a cross of *pid-TD1* and *np1-2* to identify plants with either *np1-2 pid-TD1*^{+/-} or *np1-2*^{+/-} *pid-TD1* genotype to investigate the effects of gene dosage on genetic interactions between the two mutants. It was very clear that heterozygous *np1-2* did not rescue *pid-TD1* (Supplemental Figure 1 [DOI](#)). Interestingly, heterozygous *pid-TD1* further enhanced *np1-2*, which is fertile and which does not produce pin-like structures (Supplemental Figure 1 [DOI](#)). Plants with the *np1-2 pid-TD1*^{+/-} genotype were essentially sterile and produced pin-like structures (Supplemental Figure 1 [DOI](#)). Our results suggest that gene dosage effects of *NPY1* and *PIN1* on *pid* are different.

Dosage effects of NPY1 and PIN1 on *pid* are opposite - Increased NPY1 gene expression is sufficient to suppress *pid* null mutants

We placed *NPY1* gene under the control of the *UBIQUITIN 10* promoter to strongly express *NPY1* (named as *NPY1 OE*). We directly transformed the construct into a *pid-c1* segregation population through Agrobacterium-mediated transformation. Among the 84 T1 plants we genotyped, we obtained 15 homozygous *pid-c1* mutants. Interestingly, none of the *pid-c1* plants produced any pin-like inflorescences and all were fully fertile, demonstrating that increases of *NPY1* gene expression were sufficient for the suppression of *pid-c1*. We genotyped T2 progenies from two *pid-c1* heterozygous T1 plants (#68 and #83) for the presence of *pid-c1* and for *pid-c1* zygosity. We used mCherry signal, which was included in the *NPY1 OE* construct, as a proxy to determine the presence and absence of the *NPY1* transgene. For each line, we identified T2 plants without the *NPY1* transgene and without the *pid-c1* mutation (called WT-68 and WT-83, respectively). We also isolated T2 plants that contained the *NPY1* overexpression construct, but did not have the *pid-c1* mutation (called *NPY1 OE* #68 in WT, and *NPY1 OE* #83 in WT). Finally, we identified T2 plants that were *pid-c1* homozygous and that had the *NPY1* transgene (called *NPY1 OE* #68 in *pid-c1* and *NPY1 OE* #83 in *pid-c1*). These genetic materials enabled us to compare the same *NPY1 OE* transgenic event in different genetic backgrounds.

As shown in Figure 2A [DOI](#) and Supplemental Figure 2 [DOI](#), both *NPY1 OE* #68 and #83 lines rescued *pid-c1*, which lacks almost the entire *PID* coding region and which is completely sterile under our laboratory conditions (Mudgett et al., 2023 [DOI](#)). Overexpression of *NPY1* in *pid-c1* completely eliminated the development of pin-like inflorescences and led to the production of plenty of fertile siliques (Figure 2A [DOI](#) & Supplemental Figure 2 [DOI](#)). *NPY1 OE* lines had notable developmental phenotypes: *NPY1 OE* plants had reduced petiole length and had shorter stature compared to WT plants (Supplemental Figure 2 [DOI](#)). At the young adult stage, *NPY1 OE* in WT and in *pid-c1* appeared very similar, suggesting that overexpression of *NPY1* made *PID* unnecessary for plant development (Supplemental Figure 2 [DOI](#)). *NPY1 OE* lines, regardless of the presence or absence of the *pid-c1* mutation, developed plenty of flowers with normal appearance (Supplemental Figure 2 [DOI](#)). Most flowers from *NPY1 OE* in *pid-c1* plants were normal, but occasionally, some flowers had minor defects such as missing a stamen, or developing fused petals with a stamen-like structure (Supplemental Figure 2 [DOI](#)). In comparison to the suppression of *pid* null mutants by *PIN1*-GFPHDR or heterozygous *pin1*, suppression of *pid-c1* by *NPY1 OE* appeared to be more complete (Mudgett et al., 2023 [DOI](#)).

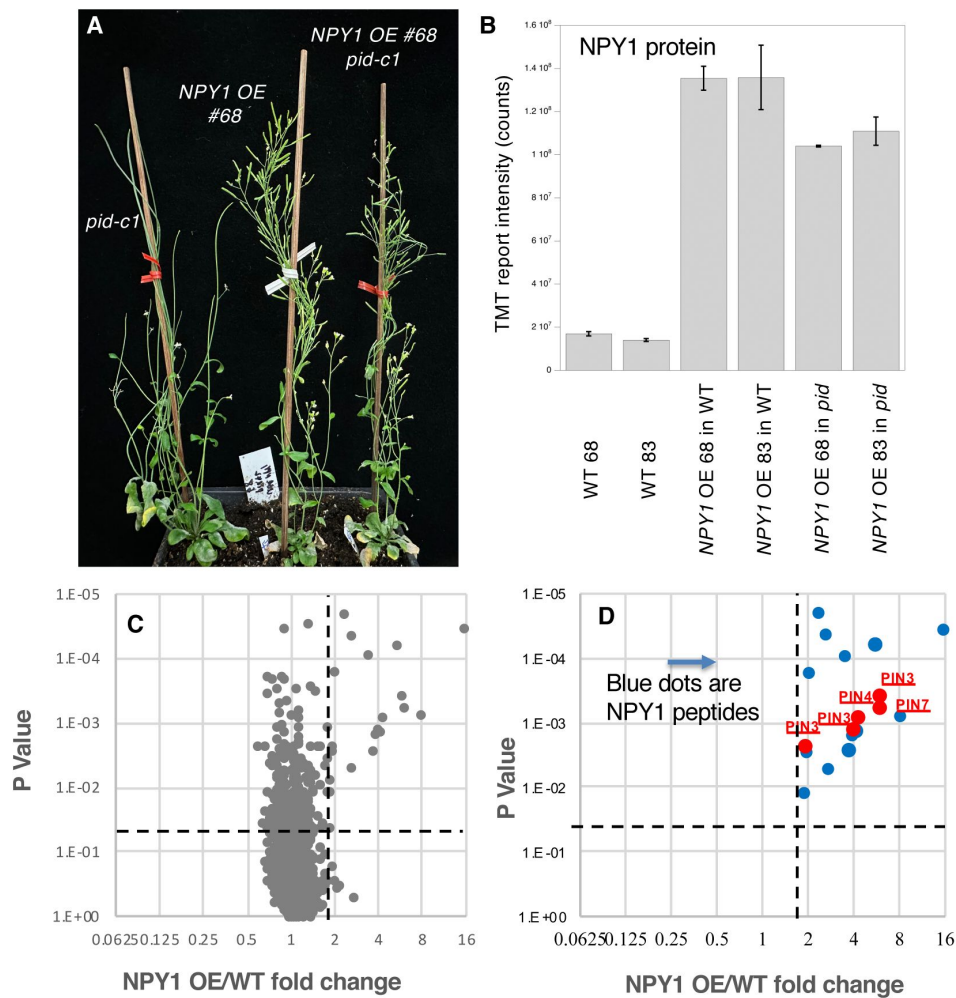


Figure 2.

Overexpression of *NPY1* rescues *pid* null mutants and triggers phosphorylation of PIN proteins.

A) The *NPY1* overexpression (OE) line #68 rescues the *pid-c1*, a null allele. Note that *pid-c1* makes pin-like inflorescences and is completely sterile. *pid-c1* no longer makes pin-like structures and is completely fertile when *NPY1* protein levels are increased. A second *NPY1* overexpression line (#83) also restored the fertility of *pid-c1* (Supplemental Figure S2). B) *NPY1* protein levels in the *NPY1 OE* lines in WT background and in *pid* mutants. In line #68, *NPY1* protein level is 8.0-fold higher than WT while line #83 has 9.7-fold higher *NPY1* protein level than WT. In the absence of *PID*, overexpression of *NPY1* leads to slightly lower *NPY1* protein levels than the same transgenic event in WT background. C) Overexpression of *NPY1* leads to an increase of phospho-peptides. The volcano plot shows the fold changes (Log2 scale) of phospho-peptide levels in *NPY1 OE* lines over WT. Data above the horizontal dotted line are statistically significant. The vertical dotted line marks 2-fold change. D) Overexpression of *NPY1* leads to phosphorylation of PIN proteins. Among the phospho-peptides that displayed the most differences between *NPY1 OE* lines and WT are peptides from *NPY1* and PIN proteins. In fact, in the right up-Quadrant in C, the only peptides were from *NPY1* and PIN proteins.

We also transformed the *NPY1 OE* construct into *pid-TD1* and *pid-TD2*, which are T-DNA insertional null alleles (Mudgett et al., 2023 [↗](#)). Both *pid* T-DNA mutants were rescued by overexpression of *NPY1*, demonstrating that *NPY1 OE* could rescue different types of *pid* mutants (Supplemental Figure 3 [↗](#)). Some of the siliques in the rescued *pid-TD* plants had only one valve, a phenotype that was likely caused by a partial suppression of *pid-TD* (Supplemental Figure 3 [↗](#)). Our results demonstrated that overexpression of *NPY1* rescued *pid* mutants and the suppression of *pid* by *NPY1 OE* was not caused by any background mutations, a particular T-DNA insertion site, or dependence on special *pid* mutations. Our results clearly showed that *NPY1* and *PIN1* had opposite dosage effects on *pid* mutants.

NPY1 protein accumulates less in the absence of PID

We conducted proteomic analysis to determine the expression levels of *NPY1* in the transgenic lines (Figure 2B [↗](#)). In *NPY1 OE* line #68 in WT, *NPY1* protein level increased 8.0-fold compared to non-transgenic WT-68 (Figure 2B [↗](#)). The *NPY1* protein level was slightly higher in line #83 than in line #68 (9.7-fold vs 8.0-fold) (Figure 2B [↗](#)). The same *NPY1* transgenic events also led to elevated *NPY1* protein levels in *pid-c1* compared to non-transgenic WT (Figure 2B [↗](#)). Interestingly, *NPY1* protein levels in the *NPY1 OE* in *pid-c1* plants were significantly lower than *NPY1 OE* in WT. For line #68, *NPY1* level was 23% lower in *pid-c1* than in *NPY1 OE* in WT. In line #83, *NPY1* protein level was 18.5% lower in *pid-c1* than in WT. Our results suggest that PID likely plays a role in regulating *NPY1* protein accumulation.

NPY1 is phosphorylated and the phosphorylation of NPY1 does not require PID

We identified 15 phospho-peptide of *NPY1* (Supplemental Table 1 [↗](#)). Among the phosphorylation sites identified, most of them were located in the peptide ANHSPVASVAASSHSPVEK, which has five serine residues and which is located immediately downstream of the conserved NPH3 domain (Supplemental Figure 4 [↗](#)). Previous studies have identified two phosphorylation sites in *NPY1*: S514 and S553 (Matthes et al., 2024 [↗](#)). We also detected that S553 was phosphorylated (Supplemental Table 1 [↗](#) and Supplemental Figure 4 [↗](#)). Moreover, we identified a couple of phospho-serine residues outside of the C-terminal domain (Supplemental Figure 4 [↗](#)). Interestingly, S181 is located in a highly conserved WSYT motif and is located in the region between the BTB domain and the NPH3 domain (Supplemental Figure 4 [↗](#)). The S181 residue is conserved among all *NPY* proteins, but in *NPH3*, the serine residue is replaced with an alanine (Supplemental Figure 4 [↗](#)).

In the absence of PID, all of the 15 phospho-peptide of *NPY1* were still phosphorylated (Supplemental Table 1 [↗](#)). Quantitatively, all of the peptides except LHEASVK had lower phosphorylation levels in *NPY1 OE* in *pid* plants than in *NPY1 OE* in WT plants. But given that in *pid*, *NPY1* protein concentrations were about 20% lower than in WT (Figure 2B [↗](#)), the observed differences in phosphorylation levels could be caused by a decrease of *NPY1* protein concentration in *pid*.

Overexpression of NPY1 leads to increases in phosphorylation of PIN proteins

Overexpression of *NPY1* caused significant changes in the phospho-proteome (Figure 2C [↗](#)). Among the phospho-peptides that were highly enriched in *NPY1 OE* lines compared to WT, the majority of them were *NPY1* peptides (Figure 2D [↗](#)). Interestingly, the next highly enriched peptides in *NPY1 OE* lines were from PIN proteins (Figure 2D [↗](#)).

The phospho-peptides AGLQVDNGANEQVGKsDQGGAK from PIN7 and AGLNVFGGAPDNDQGGRsDQGAK from PIN3 were about 6-fold higher in *NPY1 OE* lines than in WT (Table 1 [↗](#)). The two peptides mapped to the same region of the two PIN proteins (Supplemental

Figure 5 [↗](#)). Phospho-peptides in PIN1 and PIN4 were also up-regulated in *NPY1 OE* lines compared to WT (**Table 1** [↗](#)). We did not detect phospho-peptides from PIN2, probably because PIN2 is not expressed in flowers. The two phosphorylated sites in PIN1 (S271 and S282) are conserved within the PIN family and have been previously identified and characterized (**Supplemental Figure 5** [↗](#)). Two of the PIN7 phosphorylation sites were highly conserved while the majority of the phosphorylation sites in PIN3, PIN4, and PIN7 were located in the less conserved regions of their respective hydrophilic loops (**Supplemental Figure 5** [↗](#)).

The absence of *PID* hardly affected PIN phosphorylation triggered by *NPY1 OE* (**Table 1** [↗](#)), which was consistent with our previous analysis that phosphorylation levels of PIN proteins in *pid* PIN1-GFPHDR plants were not affected compared to WT plants (Mudgett et al., 2023 [↗](#)). We noticed that previous studies uncovered four highly conserved phosphorylation sites in PIN proteins, which are named as S1, S2, S3, and S4 sites and which correspond to S231, S252, S290, and S271 in PIN1, respectively (Barbosa et al., 2018 [↗](#)) (**Supplemental Figure 5** [↗](#)). Phosphorylation of S1, S2, S3, or S4 leads to activation of PIN auxin transport and changes in PIN polarity (Lanassa Bassukas et al., 2022 [↗](#)). Given that *NPY1 OE* suppressed *pid* mutants and that decreases in *PIN1* gene dosage and PIN1-GFPHDR suppressed *pid*, we concluded that phosphorylation at the newly identified sites in PIN proteins leads to the inhibition of PIN activities.

The *NPY1* C-terminal domain is required for the suppression *pid* mutants

The *NPY1* C-terminal domain is Ser/Thr rich (**Supplemental Figure 6** [↗](#)), suggesting that the region might be important for regulating *NPY1* activity and/or interactions between *NPY1* and its partners. The *NPY1* phospho-peptides we have identified were largely concentrated in this domain (**Supplemental Figure 4** [↗](#)). We overexpressed a truncated *NPY1* that lacked the 30 C-terminal amino acid residues (called *NPY1ΔC*), of which 50% are Ser/Thr (**Supplemental Figure 6** [↗](#)), to determine whether the C-terminal tail of *NPY1* was important for *NPY1* functions and whether overexpression of *NPY1ΔC* was sufficient for suppression of *pid* mutants. We transformed the *NPY1ΔC* construct into a *pid-TD1* segregating population. Among the T1 plants that were identified to be *pid-TD1* homozygous, none were suppressed by overexpression of *NPY1ΔC*. We observed that overexpression of *NPY1ΔC* actually enhanced the *pid-TD1* phenotypes (**Supplemental Figure 7** [↗](#)).

Overexpression of *NPY1ΔC* disrupts normal responses to auxin and gravity

Overexpression of *NPY1ΔC* caused obvious developmental phenotypes (**Figure 3** [↗](#)). The *NPY1ΔC OE* plants were smaller with epinastic and darker leaves compared to WT plants (**Figure 3A** [↗](#)). Overexpression of *NPY1ΔC* led to smaller plant stature and slower development, but *NPY1ΔC OE* plants were eventually able to set seeds (**Figure 3B** [↗](#)). We noticed that *NPY1ΔC OE* plants often developed small pin-like structures (**Figures 3C** [↗](#) & **3D**). At the seedling stage, *NPY1ΔC* overexpression roots were completely agravitropic (**Figure 3E** [↗](#)). Moreover, we observed that *NPY1ΔC OE* plants were resistant to auxin (**Figure 3F** [↗](#)). In media containing 100 nM 2,4-Dichlorophenoxyacetic acid (2,4-D), roots of WT plants stopped growing while roots of *NPY1ΔC OE* plants were able to elongate (**Figure 3F** [↗](#)). Interestingly, the elongated roots of the *NPY1ΔC OE* plants were still agravitropic (**Figure 3F** [↗](#)).

Overexpression of *NPY1ΔC* is sufficient to trigger phosphorylation of PIN proteins

NPY1ΔC protein levels were increased in the analyzed *NPY1ΔC* overexpression lines (#4 and #35) (**Figure 4A** [↗](#)). The phenotypes of *NPY1ΔC OE* lines appeared to be correlated with the expression levels of *NPY1ΔC* protein. Line #35 accumulated more *NPY1ΔC* proteins than line #4 (**Figure 4A** [↗](#)).

Peptide	Protein	<i>NPY1</i> in WT68/WT 68	<i>NPY1</i> in WT83/WT 83	Average	<i>NPY1</i> in WT/WT p- value	<i>NPY1</i> in <i>pid68</i> / <i>NPY1</i> 1 in WT 68	<i>NPY1</i> in <i>pid83</i> / <i>NPY1</i> in WT 83	average	<i>NPY1</i> in <i>pid</i> / <i>NPY1</i> in WT p-value
PSNLTGAEIYsLNTTPR	PIN7	0.910767	0.854774	0.88277	0.0127662	1.1125989	1.158728	1.13566	0.03303799
EAIEtGETVPVK	PIN7	0.927104	0.853573	0.89034	0.0704714	1.0552467	1.046953	1.0511	0.08070277
PSNLTGAEIYSLNTiPR	PIN7	0.951787	0.899093	0.92544	0.406238	1.1249642	1.00391	1.06444	0.29407892
EAIEtGEiVPVK	PIN7	0.975411	0.926035	0.95072	0.6390556	1.1383874	0.979226	1.05881	0.71282051
AGLQVDNGANEQVGKsDQGGAK	PIN7	5.51541	6.651682	6.08355	0.0043124	1.3101193	0.864393	1.08726	0.72064442
ELHMFVWGsNGSPVSDR	PIN7	0.918219	0.93853	0.92837	0.0304462	0.9595547	1.049223	1.00439	0.92951323
DVNTNQQtLPTGGK	PIN3	1.428745	1.271353	1.35005	0.0116133	0.8856713	0.948391	0.91703	0.04453962
MLVPDQsHNGETK	PIN3	1.758002	1.838732	1.79837	0.0014221	0.8721628	0.927267	0.89971	0.06765343
ELHMFVWsSNGSPVSDR	PIN3	0.956359	1.099258	1.02781	0.6267869	1.0750312	1.060511	1.06777	0.07705107
AVAHPASGDFGGEQQFsFAGK	PIN3	1.664595	1.54284	1.60372	0.0030531	0.9168185	0.931886	0.92435	0.07919847
DVNTNQQTTLPTGGKsNSHDAK	PIN3	1.206078	1.214502	1.21029	0.0059499	0.8294046	1.001475	0.91544	0.23760357
LAPNSTAALQsK	PIN3	1.615052	1.295692	1.45537	0.0142695	0.8886159	1.022037	0.95533	0.34059189
AVAHPAsGDFGGEQQFsFAGK	PIN3	2.028215	1.837636	1.93293	0.002816	1.0035134	0.866937	0.93523	0.4099271
AVAHPAsGDFGGEQQFSFAGK	PIN3	3.252098	4.731796	3.99195	0.000938	1.0927152	0.864714	0.97871	0.77684497
AGLNVFGGAPDNDQGGRsDQGAK	PIN3	5.672291	6.321004	5.99665	0.0044969	1.2176372	0.777071	0.99735	0.92826952
DVNTNQQTTLPTGGKsNSHDAK	PIN3	1.305383	1.10612	1.20575	0.1712069	0.8671461	1.185892	1.02652	0.96500644
KsGGDDIGGLDSGEGER	PIN4	1.042671	1.402068	1.22237	0.1282906	1.0428406	0.974466	1.00865	0.85331221
GGGDDIGGLDsGEGER	PIN4	3.710114	5.0168	4.36346	0.0021961	1.0736966	0.907341	0.99052	0.88663827
NSNFGPGEAFFGsK	PIN1	1.520974	1.293786	1.40738	0.0074909	0.9641699	0.963473	0.96382	0.04162644
NsNFGPGEAFFGSK	PIN1	1.154638	1.201764	1.1782	0.0050359	0.9594481	0.890138	0.92479	0.07910326

Table 1.

Overexpression of *NPY1* leads to increases of phosphorylation of PIN proteins.

NPY1 in WT 68 refers to *NPY1* overexpression (OE) line #68 in WT background while WT68 refers to plants without the *NPY1* OE construct segregated from the *NPY1* OE line #68. *NPY1* in *pid68* refers to the *NPY1* OE line #68 in the *pid-1* homozygous background. The comparisons were made on basis of the same transgenic event of integrating the *NPY1* OE construct into Arabidopsis genome. Similar nomenclature is used for line #83. Fold change over 1.5 is highlighted in orange and P-value less than 0.05 is highlighted in green.

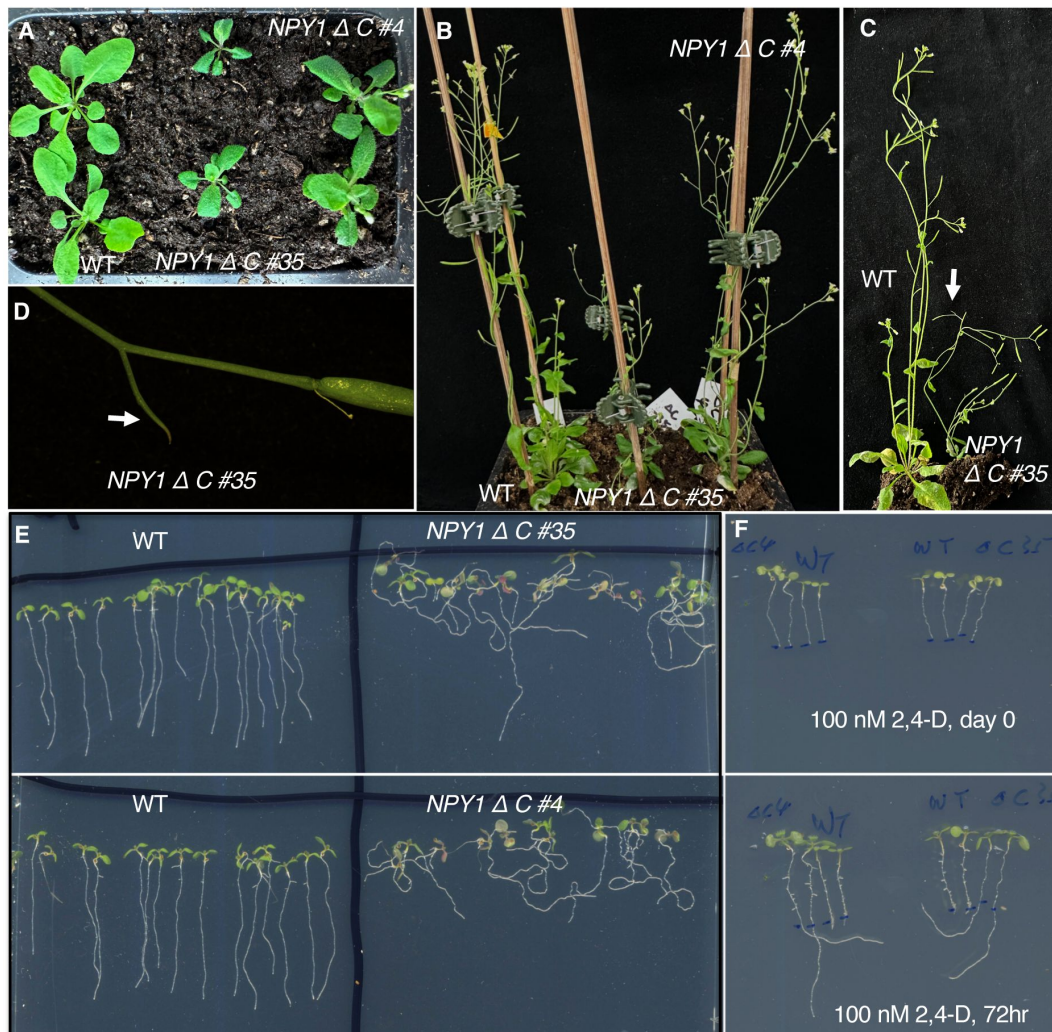


Figure 3.

Overexpression of the truncated NPY1 lacking the C-terminal 30 amino acid residues (NPY1ΔC) leads to smaller plants, agravitropic roots, and auxin resistance.

A) Overexpression of *NPY1ΔC* leads to small plant stature. The *NPY1ΔC* plants (two independent lines, #35 and #4) have short petiole and epinastic leaves. B) *NPY1ΔC* plants have smaller flowers and are slower in developing siliques. C) *NPY1ΔC* #35 is much shorter than WT. *NPY1ΔC* plants often have small pin-like structures (arrow). D) A pin-like structure (arrow) of *NPY1ΔC* #35 plants. E) *NPY1ΔC* plants have lost normal gravitropic responses. F) *NPY1ΔC* plants are auxin-resistant. Top panel, 5-day old seedlings were transferred to MS media containing 100 nM 2,4-Dichlorophenoxyacetic acid (2,4-D) and the root tips were marked. Bottom panel: The same plants from the top panel have grown for three more days. Note that roots of WT plants stopped growing, while roots of *NPY1ΔC* plants continued to grow. 2,4-D did not rescue the defects of gravitropic responses. Plants from left to right: two *NPY1ΔC* #4, two WT, two WT, two *NPY1ΔC* #35.

Line #35 had stronger phenotypes than line #4 (**Figure 3A** & B). Overexpression of *NPY1ΔC* had a profound impact on the phospho-proteome (**Figure 4B**). In the *NPY1ΔC* OE lines, we detected 18 NPY1 phospho-peptides (Supplemental Table S2, **Figure 4C**) and the majority of the sites were located in the ANHSPVASVAASSHSPVEK peptide, which was also the main phospho-peptide in the *NPY1* OE lines (**Supplemental Figure 4**, Supplemental Table S1).

Overexpression of *NPY1ΔC* also led to an increase of PIN protein phosphorylation (**Figure 4C**, Supplemental Table S2). We detected phospho-peptides from all of the long PINs (PIN1, PIN2, PIN3, PIN4, and PIN7) (Supplemental Table S2). We also detected a phospho-peptide from PIN6, which has a shorter hydrophilic loop than the long PINs (Supplemental Table S2).

Many of the identified phosphorylation sites of PIN proteins in the *NPY1ΔC* OE lines were also detected in the *NPY1* OE lines (**Supplemental Figure 8**). However, *NPY1ΔC* lines and *NPY1* OE lines also had their unique phosphorylation sites (**Supplemental Figure 8**). A caveat of this comparison is that the phospho-proteomic data were generated using different tissues. We used flowers and inflorescence heads from *NPY1* OE lines and whole seedlings of *NPY1ΔC* OE lines for the analysis. Overall, most of the phosphorylation sites were clustered to regions close to trans-membrane domain 6 (TMD6) (**Supplemental Figure 8**). Another noticeable feature is that some of the phosphorylated residues were not highly conserved among the PIN proteins (**Supplemental Figure 8**).

***NPY1 ΔC*-mediated phosphorylation of PINs correlates with phenotypes similar to those of loss-of-function *pin2* mutants**

Very intriguingly, we detected three PIN2 phospho-peptides, which were at least 2-fold more abundant in the *NPY1 ΔC* lines than in WT (**Figure 4D**, Supplemental Table S2). The level of two of the PIN2 phospho-peptides were higher in the stronger line #35 than in line #4 (**Figure 4D**). PIN2 is known to play a critical role in gravitropism and *pin2* mutants have agravitropic roots (Chen et al., 1998; Shin et al., 2005). The *NPY1ΔC* OE lines displayed complete agravitropic roots (**Figure 3E**) and developed small pin inflorescences (**Figure 3F**), suggesting that *NPY1 ΔC*-mediated phosphorylation of PINs led to their inactivation.

Discussion

In this paper, we demonstrate that *PID* becomes dispensable for flower initiation when *NPY1* is overexpressed. We further show that overexpression of *NPY1* increases phosphorylation of PIN proteins at sites mostly different from the previously characterized PIN phosphorylation sites. Phosphorylation of PIN proteins at the identified sites described here leads to the inhibition of PIN functions, rather than activation of PINs by phosphorylation at the previously identified sites (Lanassa Bassukas et al., 2022). Moreover, we show that the suppression of *pid* phenotypes by overexpressing *NPY1* requires the C-terminal motif of *NPY1*. Overexpression of *NPY1ΔC* causes a loss of gravitropism, auxin-resistance in roots, and phosphorylation of PIN proteins. Our results demonstrate that the main function of *PID* is not to directly phosphorylate PIN proteins and that *PID* affects the accumulation of *NPY1* proteins.

Our findings that overexpression of *NPY1* completely suppressed *pid* mutants demonstrate that *NPY1* functions downstream of *PID* in regulating flower initiation in Arabidopsis (**Figure 2**). Our data presented in this study are consistent with our previous hypothesis that *PID*/*NPY1*-mediated flower initiation uses a mechanism that is analogous to that of phototropism (**Figure 5**) (Cheng et al., 2007, 2008). The hypothesis was based on the observation that three signaling components in phototropism have their homologous counterparts in the flower initiation pathway (**Figure 5**). In phototropism, blue light is perceived by phototropins (PHOT1 and PHOT2), which have two LOV domains in the N-terminal part and a kinase domain in the C-terminal part (Huala

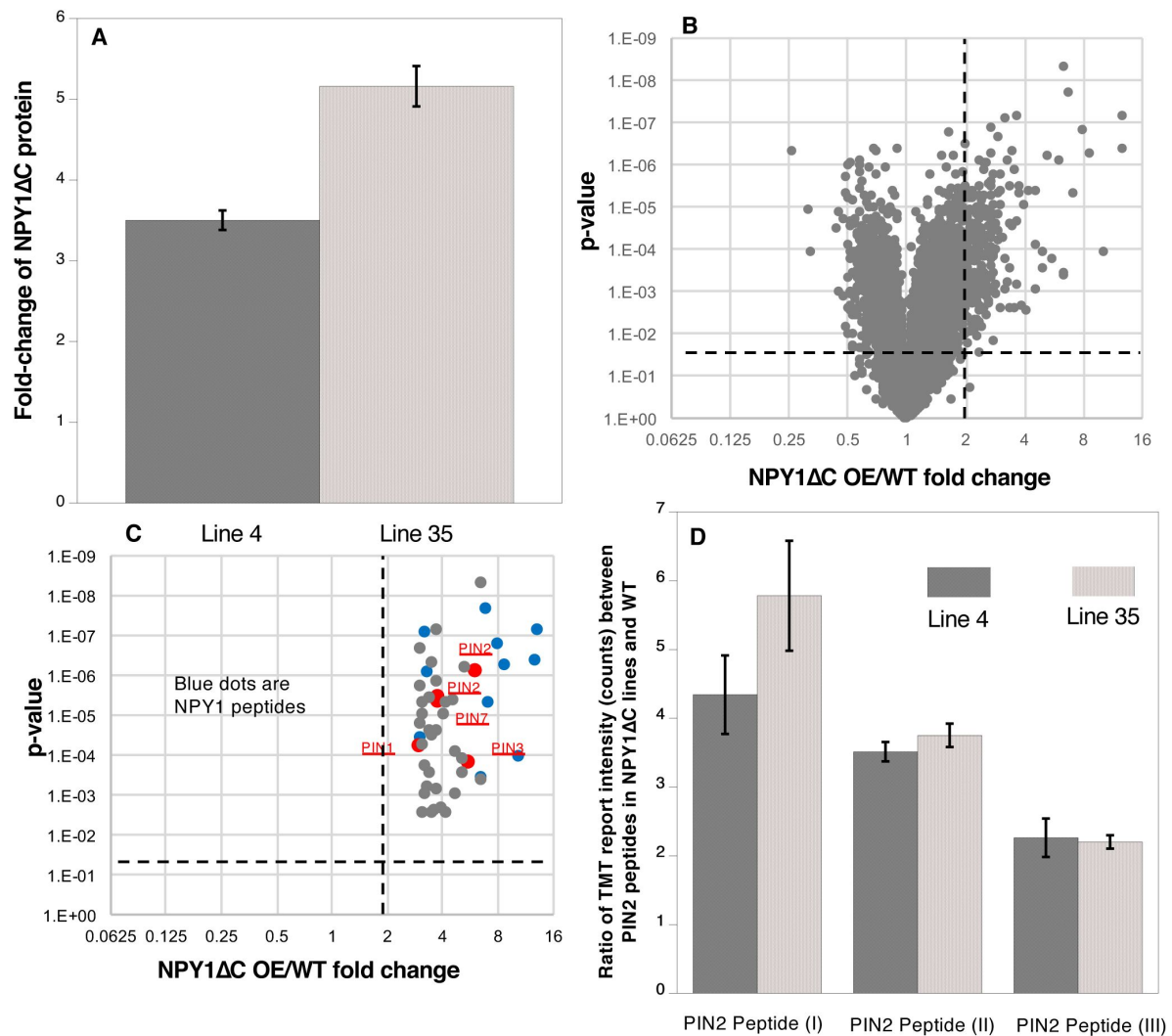


Figure 4.

Overexpression of the truncated NPY1 lacking the C-terminal 30 amino acid residues (NPY1ΔC) increases phosphorylation of PIN2 and other PINs.

A) NPY1ΔC levels are up several folds in two independent *NPY1ΔC* overexpression lines (Line 4 and Line 35). B) Overexpression of NPY1ΔC led to increases of phosphorylation of many proteins. The phospho-peptides that were at least 2-fold enriched in *NPY1ΔC* OE lines compared to WT and that are statistically significant are shown in the up-right-Quadrant. C) Among the top 50 enriched phospho-peptides in *NPY1ΔC* OE lines compared to WT, many are from NPY1 (blue dots) and PIN proteins. D) PIN2 phosphorylation increased by at least 2-fold in both line 4 and line 35 for the three detected PIN2 peptides. Peptide I is SESGGGGsGGGVGGQNK, peptide II refers to KGsDVEDGGPGPR, and peptide III is HGYTNsYGGAGAGPGGDVYSLQSSK.

et al., 1997 [↗](#); Christie et al., 1998 [↗](#); Harper et al., 2003 [↗](#)). LOV domains function as photoreceptors that regulate the kinase activities of PHOT1 and PHOT2. The kinase domain of phototropins is highly homologous to PID (Cheng et al., 2007 [↗](#)). Downstream of the phototropins is NPH3, which is homologous to NPY1 (Motchoulski and Liscum, 1999 [↗](#); Cheng et al., 2007 [↗](#)). Phototropins are the starting point of phototropic signal transduction, and are undoubtedly upstream of NPH3. Our genetic results unambiguously demonstrated that NPY1 functions downstream of PID (Figure 2 [↗](#)), further strengthening the hypothesis that phototropism and PID/NPY1 use analogous mechanisms (Figure 5 [↗](#)). In dark, NPH3 is phosphorylated and forms a complex with PHOT1 (Pedmale and Liscum, 2007 [↗](#)). Upon receiving light, NPH3 is dephosphorylated and transiently dissociated from PHOT1 (Christie et al., 2018 [↗](#); Sullivan et al., 2021 [↗](#); Haga and Sakai, 2023 [↗](#)). Continuous light exposure leads to re-constitute PHOT1-NPH3 complex and phosphorylation of NPH3 by PHOT1, suggesting that NPH3 phosphorylation by PHOT1 is rather complicated. We found that NPY1 is phosphorylated regardless of the presence of PID (Figure 2 [↗](#) and Supplemental Table 1 [↗](#)). Moreover, we identified a predominant NPY1 phospho-peptide, which is located right after the NPH3 domain (Supplemental Table 1 [↗](#) and Supplemental Figure 4). It will be interesting to determine the biological consequences of the phosphorylation sites in NPY1 through mutagenesis. At present, we do not have evidence that PID uses NPY1 as a substrate. However, it is clear that PID has an impact on NPY1 accumulation (Figure 2 [↗](#)). Without PID, NPY1 accumulates significantly less (Figure 2 [↗](#)).

Unexpectedly, overexpression of *NPY1* leads to phosphorylation of PIN proteins and likely inhibition of PIN functions. The most abundant phospho-peptides in *NPY1 OE* lines are from either NPY1 or PIN proteins (Figure 2 [↗](#)). The increased NPY1 protein levels in the *NPY1 OE* lines can partially account for the observed enrichment of phospho-peptides of NPY1 in the overexpression lines. In contrast, overall PIN protein levels were not changed in *NPY1 OE* lines compared to WT, indicating that *NPY1 OE* triggered phosphorylation of PIN proteins. Most of the phosphorylation sites in PIN proteins are not the same as those previously characterized (Lanassa Bassukas et al., 2022 [↗](#)). We hypothesize that NPY1-triggered phosphorylation of PINs actually inhibits PIN1 functions, because *NPY1 OE* suppressed *pid* (Figure 2 [↗](#)) and because *pid* is suppressed by decreasing PIN1 activity or *PIN1* gene dosage (Mudgett et al., 2023 [↗](#)). Our hypothesis is further supported by our results from overexpression of *NPY1ΔC*, which resulted in PIN2 phosphorylation (Figure 4 [↗](#)) and agravitropic roots (Figure 3 [↗](#)), a phenotype that was observed in *pin2* null mutants (Chen et al., 1998 [↗](#)). It is not clear whether *NPY1 OE*-triggered phosphorylation of PINs inhibits auxin transport activity *per se* or disrupts protein-protein interactions between PINs and their partners or both.

Both *NPY1 OE* and *NPY1ΔC* caused great changes in PIN phosphorylation and the phosphorylation led to inhibition of PIN functions (Figures 2 [↗](#) & 4). However, *NPY1 OE* suppressed *pid* mutants whereas *NPY1ΔC* could not suppress *pid* (Figure 2 [↗](#) & 3), suggesting that inhibition of PIN functions *per se* is not sufficient for suppression of *pid*. It is known that PIN and NPY proteins physically interact to form protein complexes (Glanc et al., 2021 [↗](#); Matthes et al., 2024 [↗](#)). Our results suggest that the C-terminal domain of NPY1 likely interacts with an unknown protein, which is required for normal PID and PIN functions and which needs to be recruited to the NPY-PIN complexes.

In previous models regarding the mechanisms of PID, PIN1, and NPY1, the first step is direct phosphorylation of PINs by PID (Friml et al., 2004 [↗](#); Michniewicz et al., 2007 [↗](#)). Phosphorylated PINs recruit NPY1 to the plasma membrane to maintain PIN polarity by limiting lateral diffusion of the PIN complex (Glanc et al., 2021 [↗](#)). However, that overexpression of *NPY1* eliminates the need of *PID* in flower initiation and that overexpression of *NPY1* leads to phosphorylation of PIN proteins in the absence of PID are not consistent with the model that PIN1 needs to be directly phosphorylated by PID to be activated. The fact that *pin1* and *pid* displayed opposite cotyledon/leaf phenotypes (Bennett et al., 1995 [↗](#)) and that *pid* is suppressed by heterozygous *pin1* mutants (Figure 1 [↗](#)) and PIN1-GFPHDR (Mudgett et al., 2023 [↗](#)) demonstrated that PID and PIN1 function

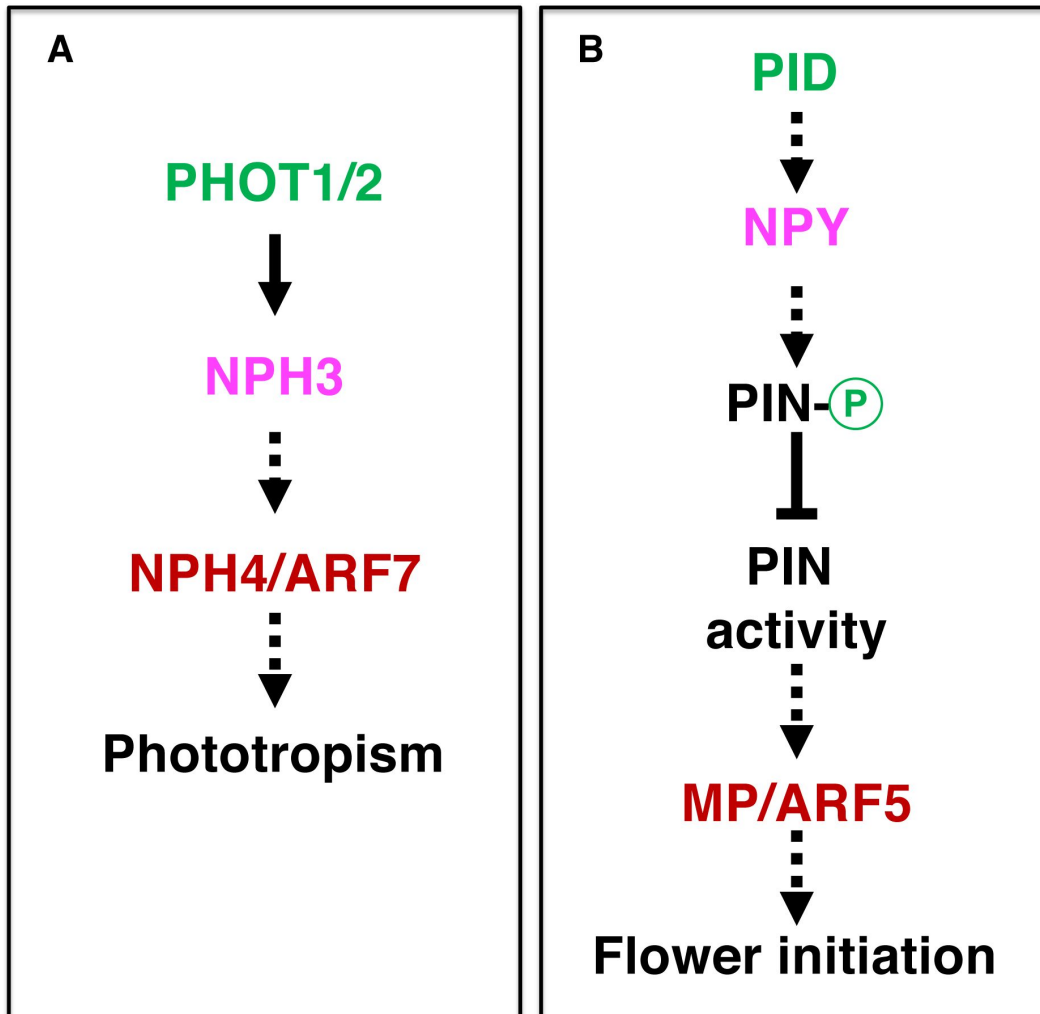


Figure 5.

The flower initiation pathway and phototropic pathway use analogous signaling mechanisms.

A) Plasma-membrane-localized phototropins perceive blue light, causing changes in phosphorylation status of NPH3 and NPH3-PHOT association. In the nucleus, transcription factor ARF7/NPH4 also plays a role. B) Pathway for auxin-mediated flower initiation. Three genes (*PID*, *NPY*, and *MP*) are homologous to their counterparts in phototropism (color coded). The dotted arrows indicate that there are gaps in our understanding of the step. PHOT: PHOTOTROPIN; NPH3: NON-PHOTOTROPICAL HYPOCOTYL 3; ARF7: AUXIN RESPONSE FACTOR 7; MP: MONOPTEROS.

in opposite directions. Our results firmly established the relative positions of PID and NPY1 in the pathway responsible for flower initiation in *Arabidopsis* (**Figure 5** [↗](#)). Moreover, we show that PIN1 functions downstream of NPY1, which recruits unknown kinase(s) to phosphorylate PIN1 and to inhibit PIN1 (**Figure 5** [↗](#)).

Materials and Methods

Plant Growth

Plants used in this study were the *Arabidopsis thaliana* Columbia-0 ecotype. Sterilized seeds were sown on a 0.7% agar-agar medium containing Murashige and Skoog basal salts at ½ concentration. Sown seeds were subjected to a two-day dark stratification period at 4°C and then moved to a growth chamber with a 16-hour day / 8-hour night cycle. After one week, seedlings were transferred to soil and maintained under the same light conditions.

Plant Transformation

Transgenic plants were created by performing the floral dipping procedure on adult plants with unopened flowers using published protocols ([Clough and Bent, 1998](#) [↗](#)). The *Agrobacterium tumefaciens* strain GV3101 was used to perform all transformations in this study. Seeds were harvested from transformed plants and transformants were selected via the expression of the mCherry fluorescent marker present on the transformed vectors ([Gao et al., 2016](#) [↗](#)).

Plasmid Construction

Vectors for gene knockouts were created by cloning two guide RNAs into the pHEE401E backbone as described in ([Mudgett et al., 2024](#) [↗](#)). Overexpression vectors were assembled via Gibson Assembly in the pHDE backbone ([Gao et al., 2016](#) [↗](#)). Primers used in this study are listed in the [Supplemental Table 3](#) [↗](#).

Proteomics Method

The following tissues were collected and were immediately frozen in liquid nitrogen for proteomic analysis. Inflorescence heads, which include flower buds and meristems from WT (Columbia background)-68, *NPY1 OE* in WT-68, *NPY1 OE* in *pid-c1*-68, WT-83, *NPY1 OE* in WT-83, *NPY1 OE* in *pid-c1*-83 were collected. Whole seedlings of 7-day old Col-WT and *NPY1ΔC* #35, and #4 were used in this study.

About 0.5 gram of frozen tissue was ground in liquid nitrogen by a mortar/pestle for 15 minutes to fine powders, and then transferred to a 50ml conical tube. Proteins were precipitated and washed by 50 ml -20 °C acetone three times, then by 50 ml -20 °C methanol three times. Samples were centrifuged at 4,000x g, 4 °C for 10 minutes. Supernatant was removed and discarded.

Protein pellets were suspended in extraction buffer (8 M Urea/100mM Tris/5mM TCEP/phosphatase inhibitors, pH 7). Proteins were first digested with Lys-C (Wako Chemicals, 125-05061) at 37 °C for 15 minutes. Protein solution was diluted 8 times to 1M urea with 100mM Tris and digested with trypsin (Roche, 03708969001) for 12 hours. Cysteines were alkylated by adding 10 mM iodoacetamide and incubating at 37 °C for 30 minutes in dark.

Digested peptides were purified on a Waters Sep-Pak C18 cartridges, eluted with 60% acetonitrile. TMT labeling ([Supplemental Table S4](#)) was performed in 60% acetonitrile/100mM Hepes, pH 7. TMT labeling efficiency was checked by LC-MS/MS to be greater than 99%. Labeled peptides from

different samples were pooled together. 150 µg of pooled peptides were analyzed by 2D-nano-LC-MS/MS for total proteome profiling and 1 mg of total peptides was used for phosphopeptide enrichment.

Phosphopeptide enrichment was performed using CeO₂ affinity capture. 20% colloidal CeO₂ (Sigma, 289744) was added to the acidified peptide solution (2% TFA/2M lactic acid/60% acetonitrile). After brief vortexing, CeO₂ with captured phosphopeptides was spun down at 5,000 g for 1 minute. Supernatant was then removed and the CeO₂ pellet was washed with 1 mL of 2% TFA/2M lactic acid/50% acetonitrile. Phosphopeptides were eluted by adding 200 µL eluting buffer (50mM (NH₄)₂HPO₄, 2M NH₃.H₂O, 10mM EDTA, pH 9.5) and vortexed briefly. CeO₂ was precipitated by adding 200 µL acetonitrile. Samples were centrifuged at 16,100 g for 1 minute. Supernatant containing phosphopeptides was removed and dried in a Speedvac. Phosphopeptides were resuspended in 100 mM citric acid and ready for mass spec analysis.

A Thermo Scientific Vanquish Neo UHPLC system (Buffer A: Water with 0.1% formic acid; Buffer B: 80% acetonitrile with 0.1% formic acid) was used to deliver a flow rate of 500 nL/min to a self-packed 3-phase capillary (360 µm OD/200 µm ID) chromatography column. Column phases were a 10 cm long reverse phase (RP1, 5 µm Zorbax SB-C18, Agilent), 6 cm long strong cation exchange (SCX, 3 µm PolySulfoethyl, PolyLC), and 20 cm long reverse phase 2 (RP2, ReproSil-Pur 120 C18-AQ, 1.9 µm), with the electrospray tip of the fused silica tubing pulled to a sharp tip.

Peptide mixtures were loaded onto RP1 using an off-line pressure chamber; and the 3 sections were joined for on-line 2D LC separation. Peptides were eluted from RP1 section to SCX section using a 0 to 80% acetonitrile gradient for 60 minutes, and then were fractionated by the SCX column section by injecting a series of ammonium acetate solutions (5 µL of 10 mM, 20 mM, 30 mM, 40 mM, 50 mM, 60 mM, 70 mM, 80 mM, 90 mM, 100 mM, and 1 M), followed by high-resolution reverse phase separation on the RP2 section of the column using an acetonitrile gradient (0–0.1 min 1% B to 5% B, 0.1–110.1 min 5% B to 35% B, 110.1–120.1 min 35% B to 60% B, 120.1–120.3 min 60% B to 95% B, 120.3–126 min 95% B).

Mass Spectra were acquired on a Thermo Exploris 480 mass spectrometer operated in positive ion mode with a source temperature of 300 °C and spray voltage of 2.1 kV. Automated data-dependent acquisition was employed of the top 20 ions with an isolation window of 0.7 Da and collision energy of 35. Precursor Fit was set to 70% of fit threshold and 0.7 Da fit window. The mass resolution is set at 100,000 for MS and 30,000 for MS/MS scans, respectively. TurboTMT was enabled. Dynamic exclusion of 30 seconds was used to improve the duty cycle.

The raw data was extracted and searched using Spectrum Mill vBI.07 (Broad Institute of MIT and Harvard). MS/MS spectra with a sequence tag length of 1 or less were considered to be poor spectra and were discarded. The remaining high-quality MS/MS spectra were searched against Arabidopsis TAIR11 protein database. A 1:1 concatenated forward-reverse database was constructed to calculate the false discovery rate (FDR). Common contaminants such as trypsin and keratin were included in the protein database. There were 96,562 protein sequences in the final protein database. Search parameters were set to Spectrum Mill's default settings with the enzyme parameter limited to full tryptic peptides with a maximum mis-cleavage of 1. Cutoff scores were dynamically assigned to each dataset to obtain the false discovery rates (FDR) of 0.1% for peptides, and 1% for proteins. Phosphorylation sites were localized to a particular amino acid within a peptide using the variable modification localization (VML) score in Spectrum Mill software. Proteins that share common peptides were grouped using principles of parsimony to address protein database redundancy. Total TMT reporter intensities were used for relative protein quantitation. Peptides shared among different protein groups were removed before TMT quantitation. Isotope impurities of TMT reagents were corrected using correction factors provided

by the manufacturer (Thermo). Median normalization was performed to normalize the protein TMT reporter intensities in which the log ratios between different TMT tags were adjusted globally such that the median log ratio was zero.

Proteomics data deposition

The raw spectra for the proteome data have been deposited in the Mass Spectrometry Interactive Virtual Environment (MassIVE) repository (massive.ucsd.edu/ProteoSAFe/static/massive.jsp, accession ID MSV000098241). FTP download link before publication: <ftp://MSV000098241@massive-ftp.ucsd.edu> ; FTP download link after publication: <ftp://massive-ftp.ucsd.edu/v10/MSV000098241> .

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Additional files

Supplementary file 1. 

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Reviewer #1 (Public review):

Summary:

The authors of this study propose a model in which NPY family regulators antagonize the activity of the *pid* mutation in the context of floral development and other auxin-related phenotypes. This is hypothesized to occur through regulation of or by PID and its action on the PIN1 auxin transporter.

Strengths:

The findings are intriguing.

Weaknesses and Major Comments:

- (1) While the findings are indeed intriguing, the mechanism of action and interaction among these components remains poorly understood. The study would benefit from significantly more thorough and focused experimental analyses to truly advance our understanding of *pid* phenotypes and the interplay among *PID*, *NPYs*, and *PIN1*.
- (2) The manuscript appears hastily assembled, with key methodological and conceptual details either missing or inconsistent. Although issues with figure formatting and clarity (e.g., lack of scale bars and inconsistent panel layout) may alone warrant revision, the content remains the central concern and must take precedence over presentation.
- (3) Given that fertile progeny are obtained from *pid*-TD *pin1*/*PIN1* and *pid* *NPY* OE lines, it would be important to analyze whether mutations and associated phenotypes are heritable. This is especially relevant since CRISPR lines can be mosaic. Comprehensive genotyping and inheritance studies are required.
- (4) The Materials and Methods section lacks essential information on how the lines were generated, genotyped, propagated, and scored. There is also generally no mention of how reproducible the observations were. These genetic experiments need to be described in detail, including the number of lines analyzed and consistency across replicates.
- (5) The nature of the *pid* alleles used in the study is not described. This is essential for interpretation.
- (6) The authors measure *PIN1* phosphorylation in response to *NPY* overexpression and conclude that the newly identified phosphorylation sites are inhibitory because they do not overlap with known activating sites. This conclusion is speculative without functional validation. Functional assays are available and must be included to substantiate this claim.
- (7) Figure 5 implies that *NPY1* acts downstream of *PID*, but there is no biochemical evidence supporting this hierarchy. Additional experiments are needed to demonstrate the epistatic or regulatory relationship.
- (8) The authors should align their genetic observations with cell biological data on *PIN1*, *PIN2*, and *PID* localization and distribution.

<https://doi.org/10.7554/eLife.108207.1.sa3>

Reviewer #2 (Public review):

Summary:

The study is well-conducted, revealing that *NPY1*, with previously less-characterized molecular functions, can suppress *pid* mutant phenotypes with a phosphorylation-based mechanism. Overexpression of *NPY1* (*NPY1*-OE) results in *PIN* phosphorylation at unique sites and bypasses the requirement for *PID* for this event. Conversely, a C-terminal deleted form of *NPY1* (*NPY1*-dC) fails to rescue *pid* despite promoting a certain phospho-profile in *PIN* proteins.

Strengths:

- (1) The careful genetic analyses of *pid* suppression by *NPY1*-OE and the inability of *NPY1*dC to do the same.

(2) Phospho-proteomics approaches reveal that NPY1-OE induces phosphorylation of PINs at non-canonical sites, independent of PID.

Weaknesses:

- (1) The native role of NPY1 is not tested by phospho-proteomics in loss-of-function *np1* mutants. Such analysis would be crucial to demonstrate that NPY1 is required for the observed phosphorylation events.
- (2) The functional consequences of the newly identified phosphorylation sites in PINs remain speculative. Site-directed mutagenesis (phospho-defective and phospho-mimetic) would help clarify their physiological roles.
- (3) The kinase responsible for NPY1-mediated phosphorylation remains unidentified. Since NPY1 is a non-kinase protein, a model involving recruitment of partner kinases (e.g., PIN-phosphorylating kinases other than PID) should be considered or discussed.

<https://doi.org/10.7554/eLife.108207.1.sa2>

Reviewer #3 (Public review):

Summary:

This manuscript from Mudgett et al. explores the relative roles of PID and NPY1 in auxin-dependent floral initiation in Arabidopsis. Micro vectorial auxin flows directed by PIN1 are essential to flower initiation, and loss of PIN1 or two of its regulators, PID and NPY1 (in a *yucca*-deficient background) phenocopies the *pin1* mutant phenotype. This group has previously shown that PID-PIN1 interactions and function are dosage-dependent. The authors pick up this thread by demonstrating that a heterozygote containing a CRISPR deletion of one copy of PIN1 can restore quasi-wild type floral initiation to *pin1*.

The authors then show that overexpression of NPY1 is sufficient to more or less restore wild-type floral initiation to the *pin1* mutant. The authors claim that this result demonstrates that NPY1 functions downstream of PID, as this ectopic abundance of NPY1 resulted in phosphorylation of PIN1 at sites that differ from sites of action of PID. The authors pursue evidence that PID action via NPY1 is analogous to the mode of action by which *phot1/2* act on NPH3 in seedling phototropism. Such a model is supported by the evidence presented herein that the C terminus of NPY1, which has abundant Ser/Thr content, is phosphorylated, and that the deletion of this domain prevents overexpression compensation of the *pin1* mutant phenotype.

While the results presented support evidence in the literature that PID acts on NPY1 to regulate PIN1 function, it is also possible that NPY1 overexpression results in limited expansion of phosphorylation targets observed with other AGC kinases. And if the *phot* model is any indication, there may be other PID targets that modulate PIN1-dependent floral initiation.

However, overexpression of the NPY1 C-terminal deletion construct resulted in phosphorylation of both PIN1 and PIN2 and agravitropic root growth similar to what is observed in *pin2* mutants. This suggests that direct PID phosphorylation of PINs and action via NPY1 can be distinguished by phosphorylation sites and by growth phenotypes.

Strengths:

A very important effort that places NPY1 downstream of PID in floral initiation.

Weaknesses:

As PID has been shown to act on sites that regulate PIN protein polarity as well as PIN protein function, it would be useful if the authors consider how their results would fit/not fit with a model where combinatorial function of NPY1 and PID regulate PIN1 in a manner similar to the way that PID appears to function combinatorially with D6PK on PIN3.

<https://doi.org/10.7554/eLife.108207.1.sa1>

Author response:

Reviewer #1 (Public review):

Summary:

The authors of this study propose a model in which NPY family regulators antagonize the activity of the pid mutation in the context of floral development and other auxin-related phenotypes. This is hypothesized to occur through regulation of or by PID and its action on the PIN1 auxin transporter.

Strengths:

The findings are intriguing.

We are pleased that the reviewer found the work interesting!

Weaknesses and Major Comments:

(1) While the findings are indeed intriguing, the mechanism of action and interaction among these components remains poorly understood. The study would benefit from significantly more thorough and focused experimental analyses to truly advance our understanding of pid phenotypes and the interplay among PID, NPYs, and PIN1.

Elucidating the mechanism of action and interaction among these components will require years of additional research. As key steps toward these goals, our work clearly established that 1) NPY1 functions downstream of PID, as overexpression of NPY1 completely suppressed pid phenotypes. This is surprising because the predominant model is that PID functions by directly phosphorylating and activating PINs without the need of NPY1 involvement. 2) In the absence of PID, NPY1 protein accumulated less in the NPY1 OE lines, suggesting that PID plays a role in affecting NPY1 stability/degradation/accumulation. We are not sure what are the exact experiments this reviewer is proposing.

Regarding pid phenotypes, pid is completely sterile in our conditions, while the suppression by NPY1 OE is very clear and the lines are fertile.

(2) The manuscript appears hastily assembled, with key methodological and conceptual details either missing or inconsistent. Although issues with figure formatting and clarity (e.g., lack of scale bars and inconsistent panel layout) may alone warrant revision, the content remains the central concern and must take precedence over presentation.

We did not include scale bars in our figures because the phenotype of interest is presence/absence of flowers. Readers should compare the mutants with the rescued plants and the WT plants.

(3) Given that fertile progeny are obtained from pid-TD pin1/PIN1 and pid NPY OE lines, it would be important to analyze whether mutations and associated phenotypes are

heritable. This is especially relevant since CRISPR lines can be mosaic. Comprehensive genotyping and inheritance studies are required.

We only use stable, heritable, Cas9-free mutants in our studies. We genotype our mutants in every generation. More details have been added to the Materials and Methods section. We provide the genetic materials we use to the scientific community when requested to enable verification and extension of our results.

(4) The Materials and Methods section lacks essential information on how the lines were generated, genotyped, propagated, and scored. There is also generally no mention of how reproducible the observations were. These genetic experiments need to be described in detail, including the number of lines analyzed and consistency across replicates.

More details have been added to the Materials and Methods section

The criticism is not fully accurate. For example, we stated in the main text: “We genotyped T2 progenies from two *pid*-c1 heterozygous T1 plants (#68 and # 83) for the presence of *pid*-c1 and for *pid*-c1 zygosity. We used mCherry signal, which was included in the NPY1 OE construct, as a proxy to determine the presence and absence of the NPY1 transgene. For each line, we identified T2 plants without the NPY1 transgene and without the *pid*-c1 mutation (called WT-68 and WT-83, respectively). We also isolated T2 plants that contained the NPY1 overexpression construct, but did not have the *pid*-c1 mutation (called NPY1 OE #68 in WT, and NPY1 OE #83 in WT). Finally, we identified T2 plants that were *pid*-c1 homozygous and that had the NPY1 transgene (called NPY1 OE #68 in *pid*-c1 and NPY1 OE #83 in *pid*-c1). These genetic materials enabled us to compare the same NPY1 OE transgenic event in different genetic backgrounds.”

The genetic materials used are freely available to the scientific community. We would like to point out that we used several *pin1* and *pid* alleles to make sure that the phenotypes are caused by the genes of interest.

(5) The nature of the pid alleles used in the study is not described. This is essential for interpretation.

The mutants were described in a previous paper (M. Mudgett, Z. Shen, X. Dai, S.P. Briggs, & Y. Zhao, Suppression of pinoid mutant phenotypes by mutations in PIN-FORMED 1 and PIN1-GFP fusion, Proc. Natl. Acad. Sci. U.S.A. 120 (48) e2312918120, <https://doi.org/10.1073/pnas.2312918120> (2023). We have added the relevant information to Materials and Methods.

(6) The authors measure PIN1 phosphorylation in response to NPY overexpression and conclude that the newly identified phosphorylation sites are inhibitory because they do not overlap with known activating sites. This conclusion is speculative without functional validation. Functional assays are available and must be included to substantiate this claim.

We concluded that the phosphorylation of PINs in NPY1 OE is inhibitory on the basis of the following: 1) *pid* is suppressed in *pin1* heterozygous backgrounds and by PIN1-GFP_{HDR}, demonstrating that partial loss of function of PIN1 or a decrease in PIN1 gene dosage, which decreases PIN1 protein expression, caused the suppression of *pid*. 2) *pid* is completely suppressed by NPY1 OE, which caused an increase of PIN phosphorylation, suggesting that phosphorylation of PINs in NPY1 OE lines is inhibitory. It is true that we do not have biochemical data to support the conclusion. We would like to point out that the phosphorylation sites in PINs identified in this work do overlap with previously identified sites.

PIN activity assays are conducted in heterologous systems that do not include NPY proteins. Since NPY is important for PIN activities, we believe that these assays may provide misleading results. Moreover, PIN1 is likely part of a large protein complex. Without knowing the composition of the complex, functional assays in heterologous systems will not be interpretable.

(7) Figure 5 implies that NPY1 acts downstream of PID, but there is no biochemical evidence supporting this hierarchy. Additional experiments are needed to demonstrate the epistatic or regulatory relationship.

We show that overexpression of NPY1 completely suppressed the *pid* phenotype, and this epistatic relationship indicates that NPY1 functions downstream of PID. Moreover, we report that PID is required for NPY1 accumulation, indicating that PID is upstream of NPY1.

(8) The authors should align their genetic observations with cell biological data on PIN1, PIN2, and PID localization and distribution.

We are hesitating in using traditional PIN1-GFP, PIN2-GFP lines, as they are not stable in our hands. Localization of PID is still not clear. We have generated PID-GFP_{HDR} lines, but we could not detect any fluorescent signals (unpublished results). In addition, maize PINOID (BIF2) localizes to the nucleus, cytoplasm and cell periphery (Skirpan, A., Wu, X. and McSteen, P. (2008), Genetic and physical interaction suggest that BARREN STALK1 is a target of BARREN INFLORESCENCE2 in maize inflorescence development. The Plant Journal, 55: 787-797. <https://doi.org/10.1111/j.1365-313X.2008.03546.x>)

We would rather wait for the proper genetic materials before devoting our effort to this.

Reviewer #2 (Public review):

Summary:

*The study is well-conducted, revealing that NPY1, with previously less-characterized molecular functions, can suppress *pid* mutant phenotypes with a phosphorylation-based mechanism. Overexpression of NPY1 (NPY1-OE) results in PIN phosphorylation at unique sites and bypasses the requirement for PID for this event. Conversely, a C-terminal deleted form of NPY1 (NPY1-dC) fails to rescue *pid* despite promoting a certain phospho-profile in PIN proteins.*

Strengths:

*(1) The careful genetic analyses of *pid* suppression by NPY1-OE and the inability of NPY1dC to do the same.*

(2) Phospho-proteomics approaches reveal that NPY1-OE induces phosphorylation of PINs at non-canonical sites, independent of PID.

Thank you for having accurately summarized the main findings

.Weaknesses:

*(1) The native role of NPY1 is not tested by phospho-proteomics in loss-of-function *npy1* mutants. Such analysis would be crucial to demonstrate that NPY1 is required for the observed phosphorylation events.*

This is an excellent point and we agree with the reviewer that analyzing loss-of-function *npy* mutants is important. The challenge is that we need to knockout NPY1, NPY3, and NPY5 to

phenocopy *pid*. We will also need to find a way to suppress the *npv* triple mutants, which are sterile, so that we can have meaningful comparisons.

(2) The functional consequences of the newly identified phosphorylation sites in PINs remain speculative. Site-directed mutagenesis (phospho-defective and phospho-mimetic) would help clarify their physiological roles.

We agree with the reviewer on this point as well. However, this is not trivial, as we have uncovered so many phosphorylation sites.

(3) The kinase responsible for NPY1-mediated phosphorylation remains unidentified. Since NPY1 is a non-kinase protein, a model involving recruitment of partner kinases (e.g., PIN-phosphorylating kinases other than PID) should be considered or discussed.

we will add a sentence to mention D6PK and other kinases in the Discussion in the revised version. We are hoping that the kinases will come out of future forward genetic screens.

Reviewer #3 (Public review):

Summary:

*This manuscript from Mudgett et al. explores the relative roles of PID and NPY1 in auxin-dependent floral initiation in Arabidopsis. Micro vectorial auxin flows directed by PIN1 are essential to flower initiation, and loss of PIN1 or two of its regulators, PID and NPY1 (in a yucca-deficient background) phenocopies the pinformed phenotype. This group has previously shown that PID-PIN1 interactions and function are dosage-dependent. The authors pick up this thread by demonstrating that a heterozygote containing a CRISPR deletion of one copy of PIN1 can restore quasi-wild type floral initiation to *pid*.*

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*While the results presented support evidence in the literature that PID acts on NPY1 to regulate PIN1 function, it is also possible that NPY1 overexpression results in limited expansion of phosphorylation targets observed with other AGC kinases. And if the *phot* model is any indication, there may be other PID targets that modulate PIN1-dependent floral initiation.*

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Strengths:

A very important effort that places NPY1 downstream of PID in floral initiation.

We thank the reviewer for the comments.

Weaknesses:

As PID has been shown to act on sites that regulate PIN protein polarity as well as PIN protein function, it would be useful if the authors consider how their results would fit/not fit with a model where combinatorial function of NPY1 and PID regulate PIN1 in a manner similar to the way that PID appears to function combinatorially with D6PK on PIN3

We agree with the reviewer that we do not have a complete picture of how NPY, PID, PIN work together to control flower initiation. Some aspects of our results are difficult to reconcile with the model of PIN1 and PID acting in tandem, i.e., by PID directly phosphorylating and activating PIN1. Indeed, our results suggest that PIN1 and PID have opposite effects on organogenesis. For example, heterozygous *pin1* (or PIN1-GFP_{HDR}, which is presumably less active than wild type PIN1) suppresses the *pid* phenotype. Moreover, *pid* and *pin1* have opposite effects on cotyledon number and true leaf number. Mutations in *PID* lead to more cotyledons and more true leaves than WT whereas *pin1* mutants make fewer cotyledons and fewer true leaves than WT (Bennett SRM, Alvarez J, Bossinger G, Smyth DR (1995) Morphogenesis in pinoid mutants of *Arabidopsis thaliana*. The Plant Journal 8: 505-520). We have elaborated on this point in the last paragraph of the Discussion.

The genetic materials we have generated may allow us to uncover additional components in the pathway from forward genetic screens, which may eventually lead to a clear picture.

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